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If, If-else , if-else-if

Question no 1: Easy `if` Programs (1 to 5)

1. Check Positive Number:

```
#include <iostream>
using namespace std;

int main() {
    int number = 10;
    if (number > 0) {
        cout << "Positive Number";
    }
    return 0;
}
```

2. Check Equal to 10:

```
#include <iostream>
using namespace std;

int main() {
    int x = 10;
    if (x == 10) {
        cout << "Number na 10";
    }
    return 0;
}
```

3. Check Less Than 50:

```
#include <iostream>
using namespace std;

int main() {
    int val = 25;
    if (val < 50) {
        cout << "Value smol pas 50";
```

```
    }
    return 0;
}
```

4. Check Non-Zero Value:

```
#include <iostream>
using namespace std;

int main() {
    int num = 7;
    if (num != 0) {
        cout << "Number no dey zero";
    }
    return 0;
}
```

5. Check Greater Than 100:

```
#include <iostream>
using namespace std;

int main() {
    int mark = 120;
    if (mark > 100) {
        cout << "High Score";
    }
    return 0;
}
```

B. Normal `if` Programs (6 to 10)

6. Check Voting Age:

```
#include <iostream>
using namespace std;

int main() {
    int age = 20;
    if (age >= 18) {
        cout << "Yu fit vote";
    }
    return 0;
}
```

7. Check Passing Marks:

```
#include <iostream>
using namespace std;

int main() {
    int score = 55;
    if (score >= 50) {
        cout << "Yu don pass";
    }
    return 0;
}
```

8. Check Even Number:

```
#include <iostream>
using namespace std;

int main() {
    int num = 8;
    if (num % 2 == 0) {
        cout << "Even Number";
    }
    return 0;
}
```

9. Check Multiple of 5:

```
#include <iostream>
using namespace std;

int main() {
    int n = 25;
    if (n % 5 == 0) {
        cout << "Multiple of 5";
    }
    return 0;
}
```

10. Check Temperature High:

```
C++
#include <iostream>
using namespace std;

int main() {
    float temp = 38.5;
```

```
    if (temp > 37.0) {
        cout << "Fever dey";
    }
    return 0;
}
```

C. Advanced `if` Programs (11 to 15)

11. Check Uppercase Character:

C++
```
#include <iostream>
using namespace std;

int main() {
    char ch = 'G';
    if (ch >= 'A' && ch <= 'Z') {
        cout << "Capital Letter";
    }
    return 0;
}
```

12. Check Divisible by 3 and 7:

C++
```
#include <iostream>
using namespace std;

int main() {
    int a = 21;
    if (a % 3 == 0 && a % 7 == 0) {
        cout << "Divisible by 3 and 7";
    }
    return 0;
}
```

13. Check Leap Year (Basic):

C++
```
#include <iostream>
using namespace std;

int main() {
    int year = 2024;
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
```

```
        cout << "Leap Year";
    }
    return 0;
}
```

14. Check Range (10 to 50):

```
C++
#include <iostream>
using namespace std;

int main() {
    int x = 30;
    if (x >= 10 && x <= 50) {
        cout << "In Range";
    }
    return 0;
}
```

15. Check Triangle Validity:

```
C++
#include <iostream>
using namespace std;

int main() {
    int a = 60, b = 60, c = 60;
    if (a + b + c == 180) {
        cout << "Valid Triangle";
    }
    return 0;
}
```

SECTION 2: `if-else` Programs (15 Total)

A. Easy `if-else` Programs (16 to 20)

16. Positive or Negative:

```
C++
#include <iostream>
using namespace std;

int main() {
```

```
    int n = -5;
    if (n >= 0) {
        cout << "Positive";
    } else {
        cout << "Negative";
    }
    return 0;
}
```

17. Greater of Two Numbers:

C++

```
#include <iostream>
using namespace std;

int main() {
    int a = 15, b = 20;
    if (a > b) {
        cout << "A Big";
    } else {
        cout << "B Big";
    }
    return 0;
}
```

18. Equal or Not Equal:

C++

```
#include <iostream>
using namespace std;

int main() {
    int x = 5, y = 5;
    if (x == y) {
        cout << "Equal";
    } else {
        cout << "Not Equal";
    }
    return 0;
}
```

19. Check Zero or Non-Zero:

C++

```
#include <iostream>
using namespace std;
```

```
int main() {
    int val = 0;
    if (val == 0) {
        cout << "Zero";
    } else {
        cout << "Not Zero";
    }
    return 0;
}
```

20. Check Pass/Fail:

C++

```
#include <iostream>
using namespace std;

int main() {
    int marks = 33;
    if (marks >= 40) {
        cout << "Pass";
    } else {
        cout << "Fail";
    }
    return 0;
}
```

B. Normal `if-else` Programs (21 to 25)

21. Odd or Even:

C++

```
#include <iostream>
using namespace std;

int main() {
    int num = 13;
    if (num % 2 == 0) {
        cout << "Even";
    } else {
        cout << "Odd";
    }
    return 0;
}
```

22. Eligible for Driving License:

C++

```
#include <iostream>
using namespace std;

int main() {
    int age = 16;
    if (age >= 18) {
        cout << "Fit Drive";
    } else {
        cout << "No Fit Drive";
    }
    return 0;
}
```

23. Maximum of Two Quantities:

C++

```
#include <iostream>
using namespace std;

int main() {
    double price1 = 99.5, price2 = 120.0;
    if (price1 > price2) {
        cout << "Price 1 High";
    } else {
        cout << "Price 2 High";
    }
    return 0;
}
```

24. Check Divisibility by 10:

C++

```
#include <iostream>
using namespace std;

int main() {
    int n = 45;
    if (n % 10 == 0) {
        cout << "Divisible by 10";
    } else {
        cout << "Not Divisible by 10";
    }
    return 0;
```

```
}
```

25. Vowel or Consonant (Simple):

```
C++
#include <iostream>
using namespace std;

int main() {
    char ch = 'a';
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
        cout << "Vowel";
    } else {
        cout << "Consonant";
    }
    return 0;
}
```

C. Advanced `if-else` Programs (26 to 30)

26. Leap Year Check:

```
C++
#include <iostream>
using namespace std;

int main() {
    int y = 2023;
    if ((y % 4 == 0 && y % 100 != 0) || (y % 400 == 0)) {
        cout << "Leap Year";
    } else {
        cout << "Not Leap Year";
    }
    return 0;
}
```

27. Alphabet or Digit:

```
C++
#include <iostream>
using namespace std;

int main() {
    char ch = '9';
    if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
```

```
        cout << "Alphabet";
    } else {
        cout << "Not Alphabet";
    }
    return 0;
}
```

28. Discount Eligibility:

```
C++
#include <iostream>
using namespace std;

int main() {
    int amount = 1200;
    if (amount >= 1000) {
        cout << "Discount Granted";
    } else {
        cout << "No Discount";
    }
    return 0;
}
```

29. Check Valid Password Length:

```
C++
#include <iostream>
using namespace std;

int main() {
    int len = 7;
    if (len >= 8) {
        cout << "Strong Password";
    } else {
        cout << "Weak Password";
    }
    return 0;
}
```

30. Quadrant Check (Basic):

```
C++
#include <iostream>
using namespace std;

int main() {
```

```
    int x = 5, y = -2;
    if (x > 0 && y > 0) {
        cout << "First Quadrant";
    } else {
        cout << "Other Quadrant";
    }
    return 0;
}
```

SECTION 3: `if-else-if` Programs (15 Total)

A. Easy `if-else-if` Programs (31 to 35)

31. Positive, Negative, or Zero:

C++
```
#include <iostream>
using namespace std;

int main() {
    int num = 0;
    if (num > 0) {
        cout << "Positive";
    } else if (num < 0) {
        cout << "Negative";
    } else {
        cout << "Zero";
    }
    return 0;
}
```

32. Simple Grading (3 Levels):

C++
```
#include <iostream>
using namespace std;

int main() {
    int marks = 75;
    if (marks >= 80) {
        cout << "Grade A";
```

```
    } else if (marks >= 50) {
        cout << "Grade B";
    } else {
        cout << "Grade C";
    }
    return 0;
}
```

33. Compare Two Numbers (3 Cases):

C++

```
#include <iostream>
using namespace std;

int main() {
    int a = 10, b = 20;
    if (a > b) {
        cout << "A Big";
    } else if (b > a) {
        cout << "B Big";
    } else {
        cout << "Both Equal";
    }
    return 0;
}
```

34. Speed Check:

C++

```
#include <iostream>
using namespace std;

int main() {
    int speed = 80;
    if (speed > 100) {
        cout << "Too Fast";
    } else if (speed >= 40) {
        cout << "Normal Speed";
    } else {
        cout << "Slow";
    }
    return 0;
}
```

35. Number Size Category:

C++
```
#include <iostream>
using namespace std;

int main() {
    int x = 500;
    if (x < 10) {
        cout << "1 Digit";
    } else if (x < 100) {
        cout << "2 Digits";
    } else {
        cout << "3 or More Digits";
    }
    return 0;
}
```

B. Normal `if-else-if` Programs (36 to 40)

36. Weekday Name (1-3):

C++
```
#include <iostream>
using namespace std;

int main() {
    int day = 2;
    if (day == 1) {
        cout << "Monday";
    } else if (day == 2) {
        cout << "Tuesday";
    } else if (day == 3) {
        cout << "Wednesday";
    } else {
        cout << "Other Day";
    }
    return 0;
}
```

37. Age Group Classification:

C++
```
#include <iostream>
using namespace std;

int main() {
    int age = 25;
```

```
    if (age < 13) {
        cout << "Child";
    } else if (age < 20) {
        cout << "Teenager";
    } else if (age < 60) {
        cout << "Adult";
    } else {
        cout << "Senior";
    }
    return 0;
}
```

38. Temperature Level:

C++

```
#include <iostream>
using namespace std;

int main() {
    int t = 32;
    if (t > 40) {
        cout << "Very Hot";
    } else if (t >= 20) {
        cout << "Warm";
    } else if (t >= 0) {
        cout << "Cold";
    } else {
        cout << "Freezing";
    }
    return 0;
}
```

39. Largest of Three Numbers:

C++

```
#include <iostream>
using namespace std;

int main() {
    int a = 12, b = 25, c = 9;
    if (a >= b && a >= c) {
        cout << "A Largest";
    } else if (b >= a && b >= c) {
        cout << "B Largest";
    } else {
        cout << "C Largest";
```

```
    }
    return 0;
}
```

40. Simple Calculator Operation Selection:

```
C++
#include <iostream>
using namespace std;

int main() {
    char op = '+';
    int x = 10, y = 5;
    if (op == '+') {
        cout << x + y;
    } else if (op == '-') {
        cout << x - y;
    } else if (op == '*') {
        cout << x * y;
    } else {
        cout << "Invalid Operator";
    }
    return 0;
}
```

C. Advanced `if-else-if` Programs (41 to 45)

41. Complete Student Grading System:

```
C++
#include <iostream>
using namespace std;

int main() {
    int score = 87;
    if (score >= 90) {
        cout << "Grade A+";
    } else if (score >= 80) {
        cout << "Grade A";
    } else if (score >= 70) {
        cout << "Grade B";
    } else if (score >= 60) {
        cout << "Grade C";
    } else if (score >= 50) {
        cout << "Grade D";
    } else {
```

```
        cout << "Grade F";
    }
    return 0;
}
```

42. Electricity Bill Calculation:

C++

```
#include <iostream>
using namespace std;

int main() {
    int units = 250;
    if (units <= 100) {
        cout << "Bill: " << units * 5;
    } else if (units <= 200) {
        cout << "Bill: " << units * 8;
    } else if (units <= 300) {
        cout << "Bill: " << units * 10;
    } else {
        cout << "Bill: " << units * 12;
    }
    return 0;
}
```

43. Character Type Classifier:

C++

```
#include <iostream>
using namespace std;

int main() {
    char ch = '#';
    if (ch >= 'A' && ch <= 'Z') {
        cout << "Uppercase Letter";
    } else if (ch >= 'a' && ch <= 'z') {
        cout << "Lowercase Letter";
    } else if (ch >= '0' && ch <= '9') {
        cout << "Digit";
    } else {
        cout << "Special Character";
    }
    return 0;
}
```

44. BMI Calculator Category:

C++

```
#include <iostream>
using namespace std;

int main() {
    float bmi = 22.5;
    if (bmi < 18.5) {
        cout << "Underweight";
    } else if (bmi <= 24.9) {
        cout << "Normal Weight";
    } else if (bmi <= 29.9) {
        cout << "Overweight";
    } else {
        cout << "Obese";
    }
    return 0;
}
```

45. Tax Bracket Calculator:

C++

```
#include <iostream>
using namespace std;

int main() {
    double income = 650000;
    if (income <= 250000) {
        cout << "No Tax";
    } else if (income <= 500000) {
        cout << "Tax Rate 5%";
    } else if (income <= 1000000) {
        cout << "Tax Rate 20%";
    } else {
        cout << "Tax Rate 30%";
    }
    return 0;
}
```